Adobe researchers release image captioning AI model CLIP-S

CLIP-S, an open source image-captioning AI model developed by researchers at Adobe and the University of North Carolina (UNC), has been released.A paper describing the model and experiments was submitted to the 2022 Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). CLIP-S generates captions from an input image using a Transformer model. The model uses CLIP-S during training to determine how well the generated caption describes the image; this score is used as a reward signal for reinforcement learning (RL).

The team has fine-tuned CLIP-S with negative caption examples generated by randomly modifying reference captions to improve the grammar of the generated captions. To address the shortcomings of existing image-captioning evaluation methods, the team has also created FineCapEval, a new benchmark data set that includes more fine-grained image captions describing image backgrounds and object relationships.

The team used the COCO data set as a benchmark to compare the captions of their model to those of several baseline models. Although a baseline model outperformed CLIP-S on text-based metrics like BLEU, CLIP-S outperformed on image-text and text-to-image retrieval metrics. It also outperformed baselines on the team's new FineCapEval benchmark "significantly." Finally, human judges "strongly" preferred CLIP-S-generated captions over baseline-model-generated captions.

Darktable releases version 4.0

Darktable, an open source alternative to Adobe Lightroom, has released version 4.0, which includes a massive list of changes, new features, and bug fixes, including new colour science tools and a completely new user interface. Developer Pascal Obry has published a detailed changelog for Darktable 4.0, which includes a plethora of platform updates. The first noticeable difference is the addition of new colour science tools, specifically, colour and exposure mapping.

CodeSee launches Open Source Hub

Open Source Hub has been launched by CodeSee, the code visibility platform that detects, visualises, and automates code understanding for a maintainable, resilient codebase (OSH). Through open source communities worldwide, OSH provides a place for developers of all skill levels to learn, contribute, explore, and connect. Rather than being a community dedicated to highlighting projects for developers to work on, OSH, according to the company, provides developers with tools to onboard and fully understand all of the code in an open source project.

With OSH, developers can now visualise the impact of changes in the codebase by using CodeSee to create a visualisation of the codebase, as well as build personal profiles and portfolios that highlight their contribution and its impact.

Developers can also search for and start projects that meet their needs and interests, as well as access engaging content and programmes on a regular basis and participate in events such as livestreams, public stages, and hackathons.

"We need more developers learning from and contributing to open source so that all of our codebases are more maintainable and resilient. Every codebase is affected by open source, so we need to help each other ramp up in these codebases quickly and support one another right now," said Shanea Leven, co-founder and CEO at CodeSee.